

SUDO, CHATTR, ALIAS, Job Scheduling

Sudo

Sudo command allows us to run a command as another user does. Sudo stands for "super user do" . If we want to permit a normal user to execute some commands that are only run by root , we can use sudo.

Granting user kevin to run "ls" command as root,

1. Getting the full path of "ls" command

```
[root@localhost ~]# which ls
alias ls='ls --color=auto'
/usr/bin/ls
```

2. Open the /etc/sudoers, add the following line at the bottom of the file.

```
[root@localhost ~]# vim /etc/sudoers
```

```
## Same thing without a password
# %wheel      ALL=(ALL)      NOPASSWD: ALL

## Allows members of the users group to mount and unmount the
## cdrom as root
# %users      ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom

## Allows members of the users group to shutdown this system
# %users      localhost=/sbin/shutdown -h now

## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
#includedir /etc/sudoers.d
clado  ALL=(root)      /usr/bin/ls,/usr/sbin/useradd,/usr/bin/passwd
kevin  ALL=(root)      /usr/bin/ls
```

3. Testing the Newly granted permission.

```
[root@localhost ~]# su - kevin
[kevin@localhost ~]$ sudo ls /root/
[sudo] password for kevin:
anaconda-ks.cfg  httpd-2.4.46.tar.gz  php-7.4.2  qw.txt
Downloads        initial-setup-ks.cfg  php-7.4.2.tar.bz2  tset.txt
httpd-2.4.46     openvz.repo          Pictures
[kevin@localhost ~]$
```

4. Granting sudo permissions to execute all user administration related commands.

We are going to grant permissions to execute commands such as useradd, password to the user kevin. For This we need to find out the absolute path of each command. We use "which" command to do this.

```
[root@localhost ~]# which useradd
/usr/sbin/useradd
[root@localhost ~]# which passwd
/usr/bin/passwd
[root@localhost ~]#
```

5. Open the /etc/sudoers and add the following line and save it.

```
## Allows members of the users group to mount and unmount the
## cdrom as root
# %users  ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom

## Allows members of the users group to shutdown this system
# %users  localhost=/sbin/shutdown -h now

## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
#includedir /etc/sudoers.d
clado    ALL=(root)    /usr/bin/ls,/usr/sbin/useradd,/usr/bin/passwd
kevin    ALL=(root)    /usr/bin/ls,/usr/sbin/useradd,/usr/bin/passwd
```

Testing

```
[root@localhost ~]# su - kevin
[kevin@localhost ~]$ sudo useradd siya
[sudo] password for kevin:
[kevin@localhost ~]$ sudo passwd siya
Changing password for user siya.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[kevin@localhost ~]$
```

ALIASING

It can be used to avoid typing long commands.

Open the `/root/.bashrc` file and type the following line,

```
alias f="free"
```

Save the file and refresh it using the source command.

```
# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi
alias k="history"
alias apple="awk -F : '{print \$1}' /etc/passwd"
alias f="free"

~
~
~
~
~
~
~
~
~
~

~/bashrc [readonly] 15L 273C
```

```
[root@localhost ~]# source /root/.bashrc
[root@localhost ~]# f
              total        used        free      shared  buff/cache   available
Mem:          3702952    2372952     220712       297944     1109288     789512
Swap:              0              0              0
```

Now you will get the output of the free command by using the letter “f”.

It can be used to avoid typing long commands.

To make system wide make the changes in
Vim /etc/bashrc

To add only for root user
Vim /root/.bashrc

To add only for a user
Vim /home/username/.bashrc

To add temporary aliasing simply type the command on the terminal
#alias f="free"

To remove temporary aliasing
#unalias f

Chattr

- Changing attributes of files and directories.
- Changing the attributes of files and directories helps the root user to prevent any accidental deletion or modification of them.
- To change the attribute we use the chattr command.
- The most important attributes are
 1. immutable attribute
 2. appending attribute.

Immutable attribute

- A file with the immutable attribute cannot be modified, it cannot be deleted or renamed, no link can be created to this file and no data can be written to the file.
- Only the super user can set or clear this attribute
- We can copy the file even with the immutable attribute
- The letter "i" is used to denote the immutable flag. The "+i" will enable the immutable flag and "-i" will remove the immutable flag.

#chattr +i /root/ab.txt

```
[root@wrkstation ~]# chattr +i /course/attribute/attribute.txt
```

Now you can only read this file. All other actions except read will be denied including append, edit, rename or delete.

```
[root@wrkstation ~]# rm -rf /course/attribute/attribute.txt -----> No deletion
rm: cannot remove '/course/attribute/attribute.txt': Operation not permitted
[root@wrkstation ~]# echo "i am new line" > /course/attribute/attribute.txt --> No editing
bash: /course/attribute/attribute.txt: Permission denied
[root@wrkstation ~]# mv /course/attribute/attribute.txt newfile.txt -----> No renaming
mv: cannot remove '/course/attribute/attribute.txt': Operation not permitted
```

To show if any attribute is currently set use:

#lsattr

In case of appending attribute use

#chattr +a abc.txt

This will prevent accidental deletion and modification. But the file can be appended with new contents.

JOB SCHEDULING

Job scheduling is a feature that allows a user to submit a command for execution at a specified time in the future.

How to schedule tasks on linux

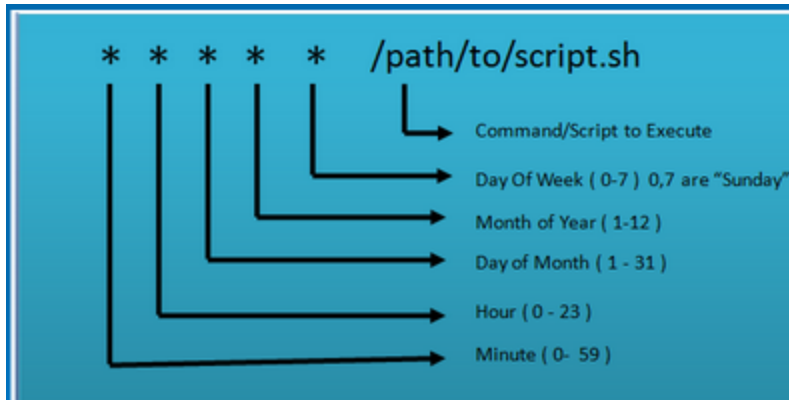
- At
- Cron
- Anacron

Difference between cron and anacron

Cron	Ancron
Minimum granularity with Cron is 1 minute	Minimum granularity with anacron is 1 day
It can be scheduled by any normal user	It can be scheduled only by the super user
expects the system to be up and running	It doesn't expect the system to be up and running all the time.
servers	desktops and laptops

Scheduling jobs with cron

Crontab : Crontab is a text file that specifies the schedule of cron jobs.



To add or create a cron job use the following command.
`#crontab -e`

```
39 1 23 03 2 touch /we.txt
```

To edit other user,
`crontab -u username -e`
To view crontab entries for current user,
`crontab -u username -l`

How to deny users

Open `/etc/cron.deny` file and add the names that you want to deny.

Configuration file = `/etc/crontab`

Content added to the crontab will be saved in the location
`/var/spool/cron`